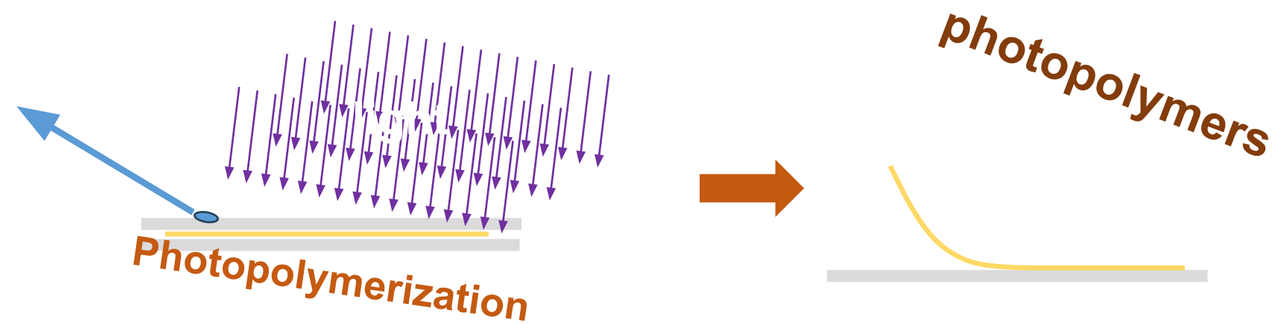
Photopolymer resins

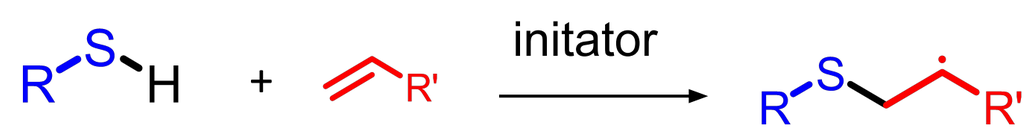
Scheme of photopolymerization

Photopolymers are a free-flowing liquid resin that hardens upon irradiation of light

Photo-curable resin:

- **Monomers / Oligomers:**
  - Form the main structure of the polymer network
  - acrylates, methacrylates, epoxies, thiol-ene
- **Photoinitiators:**
  - Initiate curing upon light exposure





Advantages

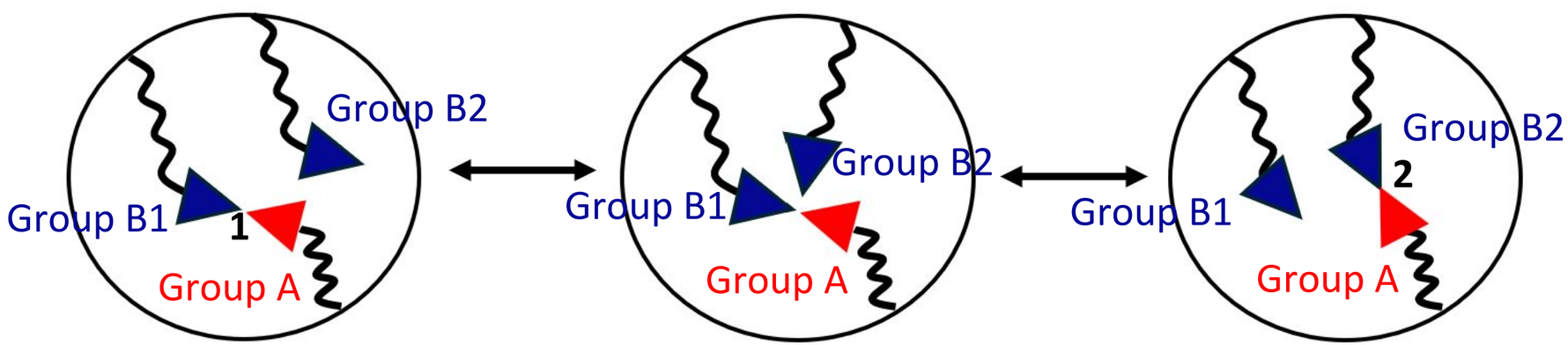
- Fast curing speed: always finishes within several minutes
- Energy efficiency

Applications: coatings, dental, and *3D printing*

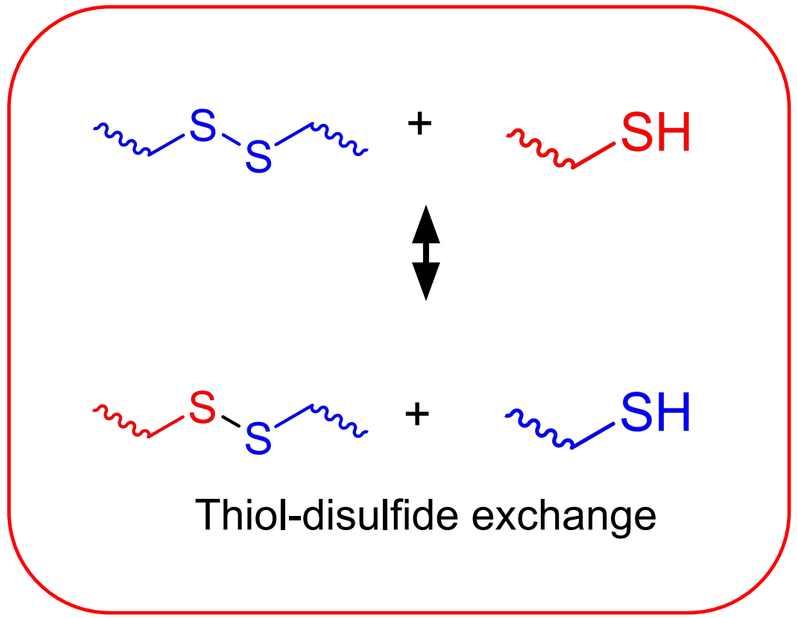
1. Kaur, M. & Srivastava, A. K. Journal of Macromolecular Science, Part C 42, 481–512 (2002).

Dynamic Covalent Chemistry (DCC): recyclability and sustainability

- Reversible/exchange covalent bonds
- Allows for structural rearrangement (i.e., *bond level*) under external stimuli: PH, heat, UV light



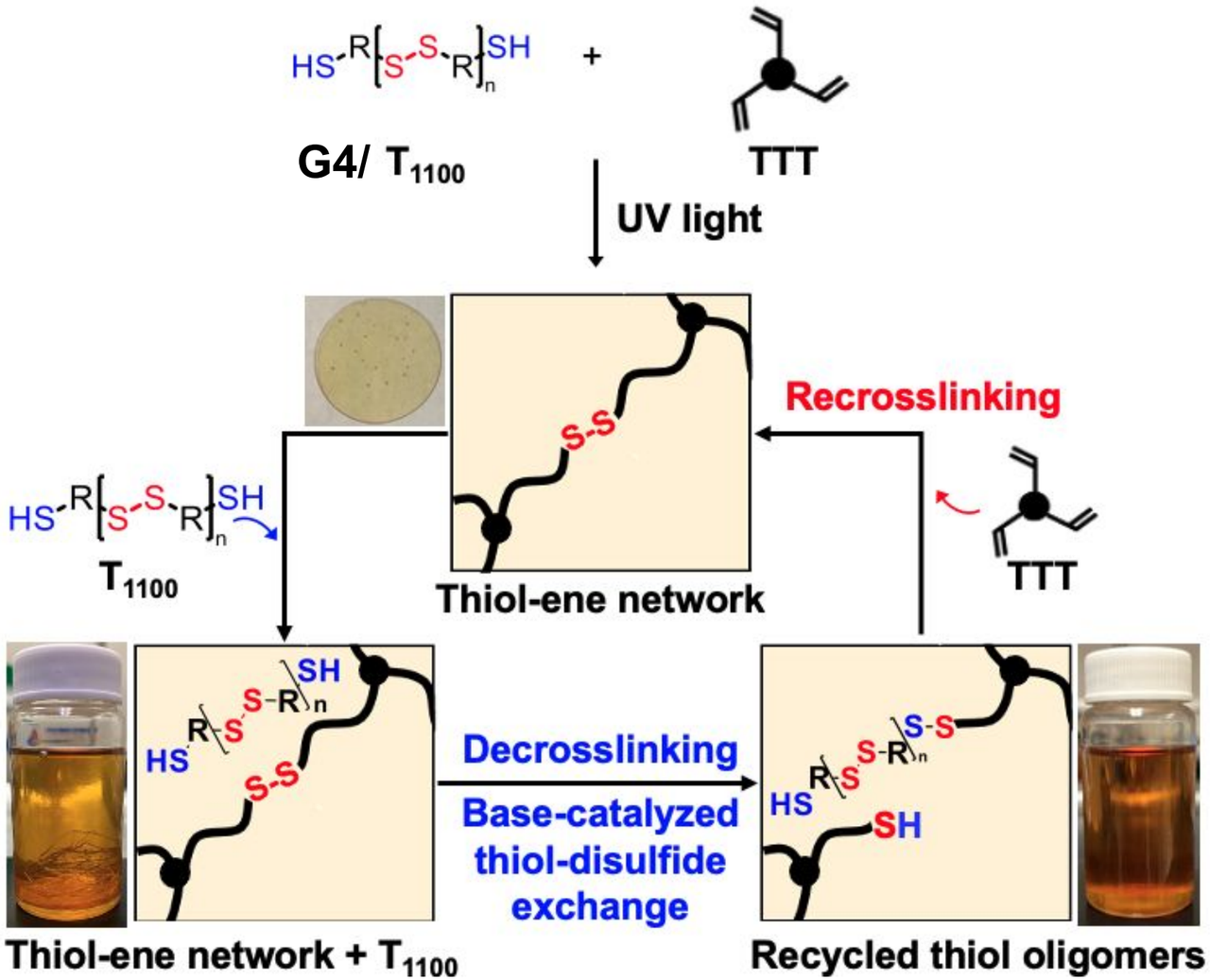
Example:

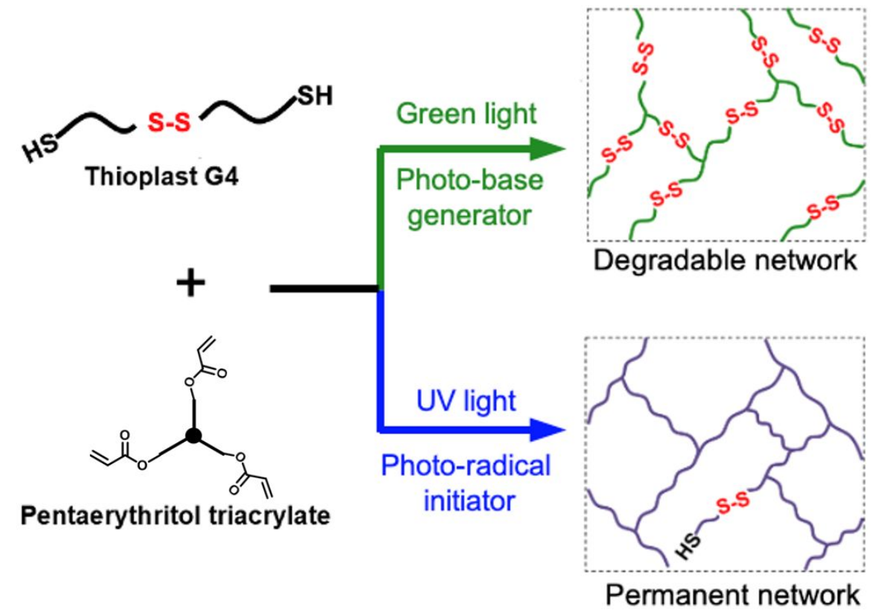


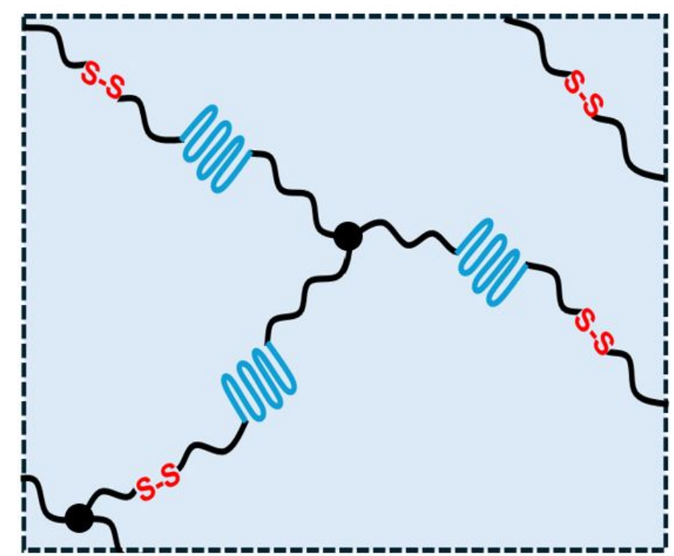
1 . Alfarhan, S. et al. Journal of Polymer Science 60, 3379–3390 (2022).

Relative work on thiol-ene dynamic covalent chemistry

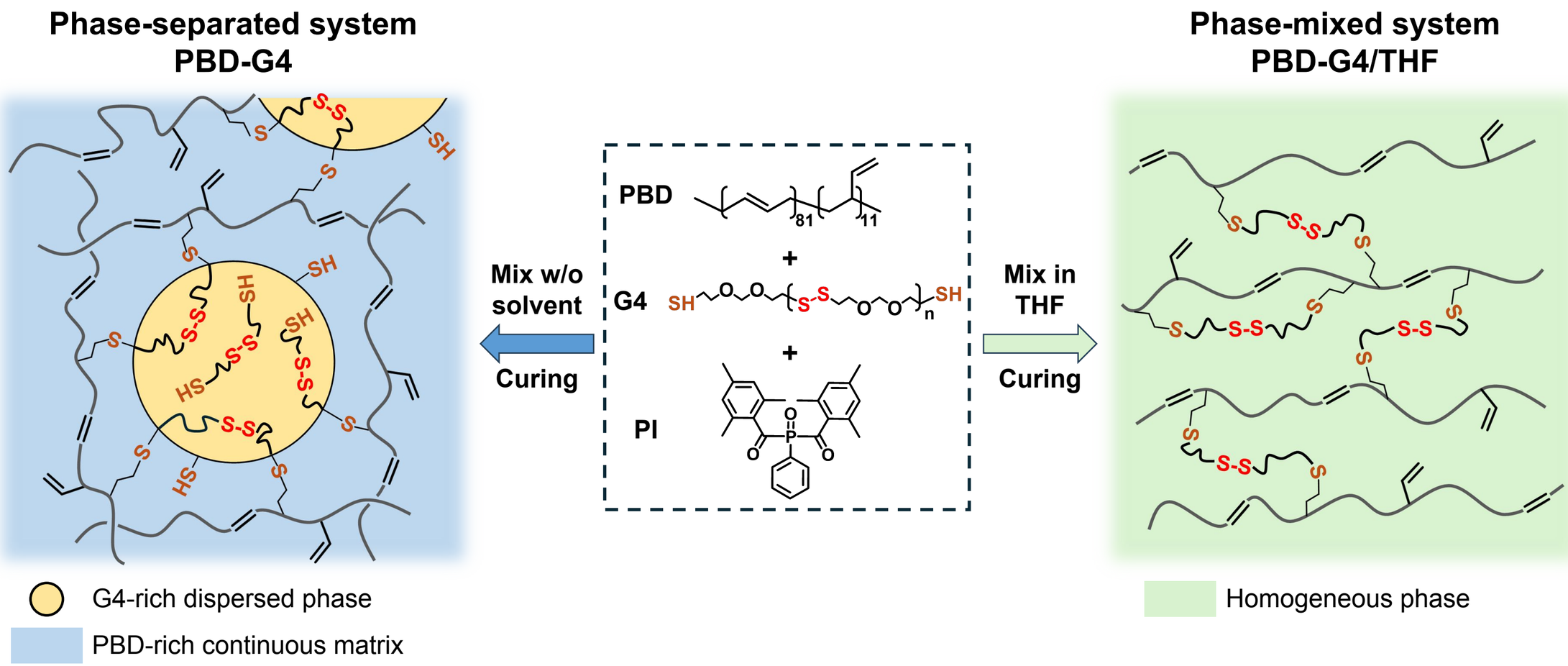
Combine the photo-curability and dynamic chemistry







1. Alfarhan, S. et al. Journal of Polymer Science 60, 3379–3390 (2022).
2. Alfarhan, S. et al. Polymer Chemistry 15, 1141–1151 (2024).
3. Alfarhan, S. et al. Small n/a, 2505122.



Scheme 1: Synthesis of phase-separated vs. phase-mixed crosslinked thiol-ene networks containing dynamic disulfide bonds.

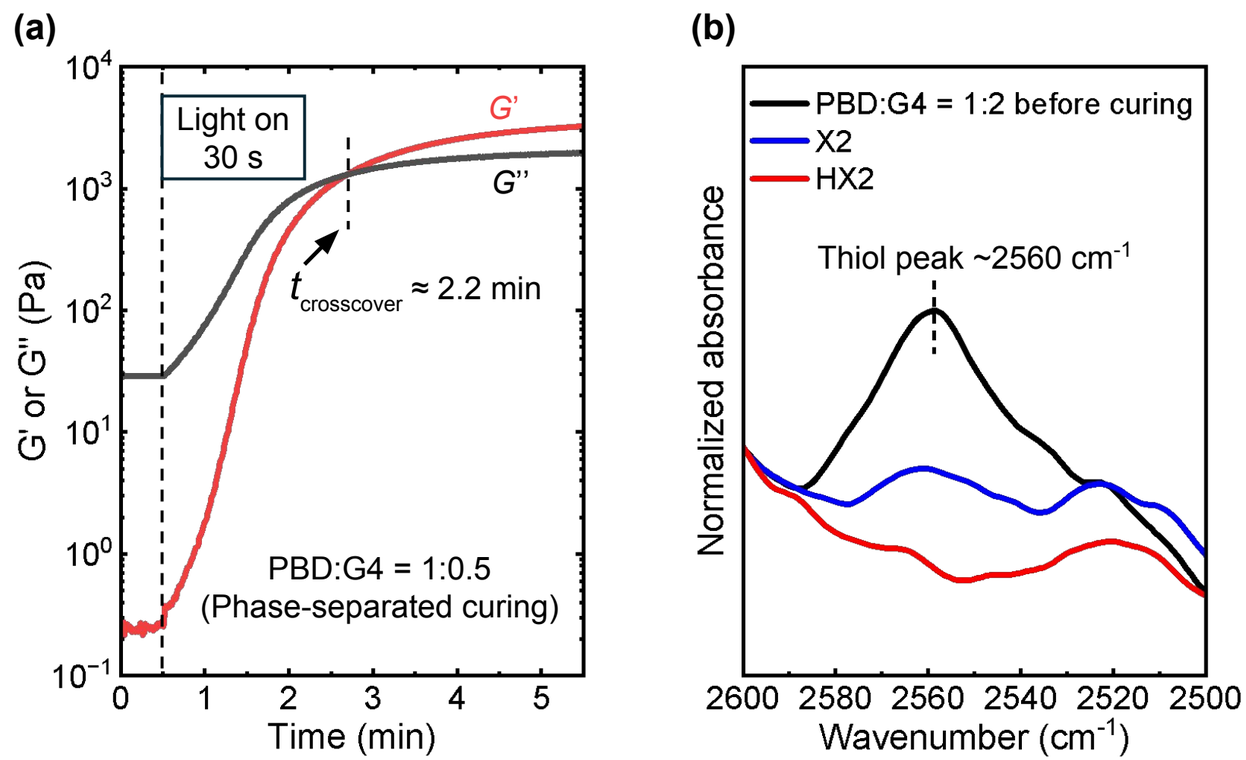


Figure 1: (a) Photo-rheological analysis of the phase-separated PBD:G4 (1:0.5) resin formulation under UV irradiation (i.e., 10 mW/cm^2), showing the evolution of G' and G'' over time; **(b)** ATR-FTIR spectra of the original PBD:G4 (1:2) resin before curing (black) and photocured X2 (blue) and HX2 (red) samples.

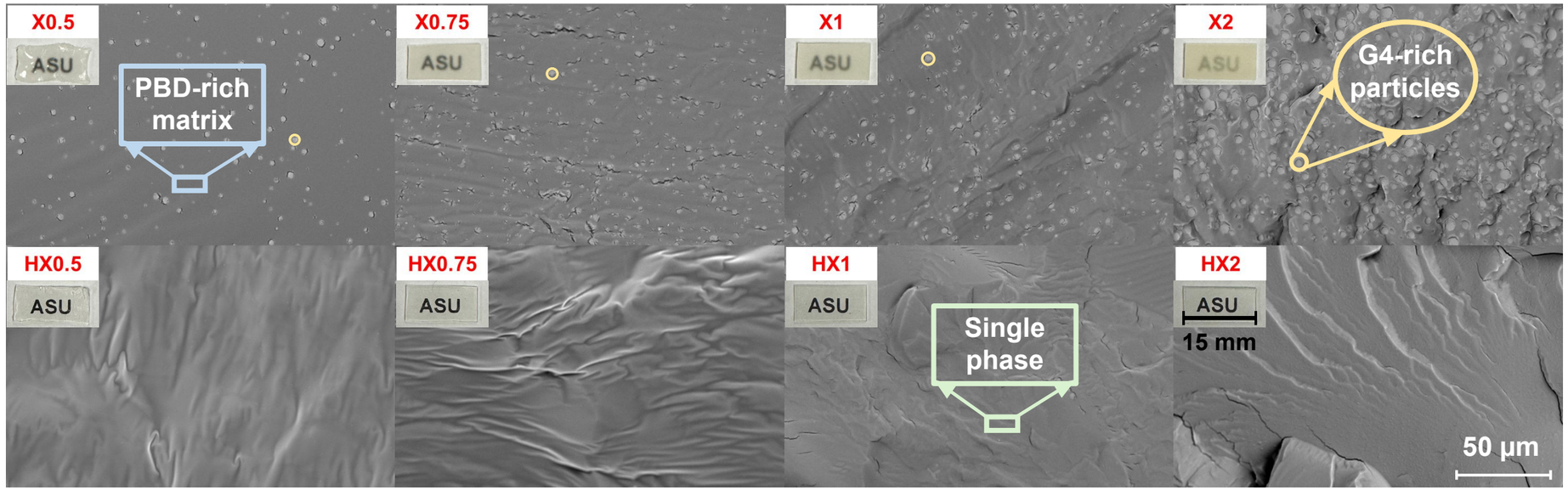


Figure 2: Representative digital photographs and SEM images for photocured PBD-G4 samples at varying compositions, with (X0.5-X2) vs. without (HX0.5-HX2) macroscopic phase separation. (Areas marked with yellow circles are G4-rich dispersed domains and areas marked with blue boxes are PBD-rich matrix, respectively; Areas marked with green boxes are single phase comprising PBD and G4.)

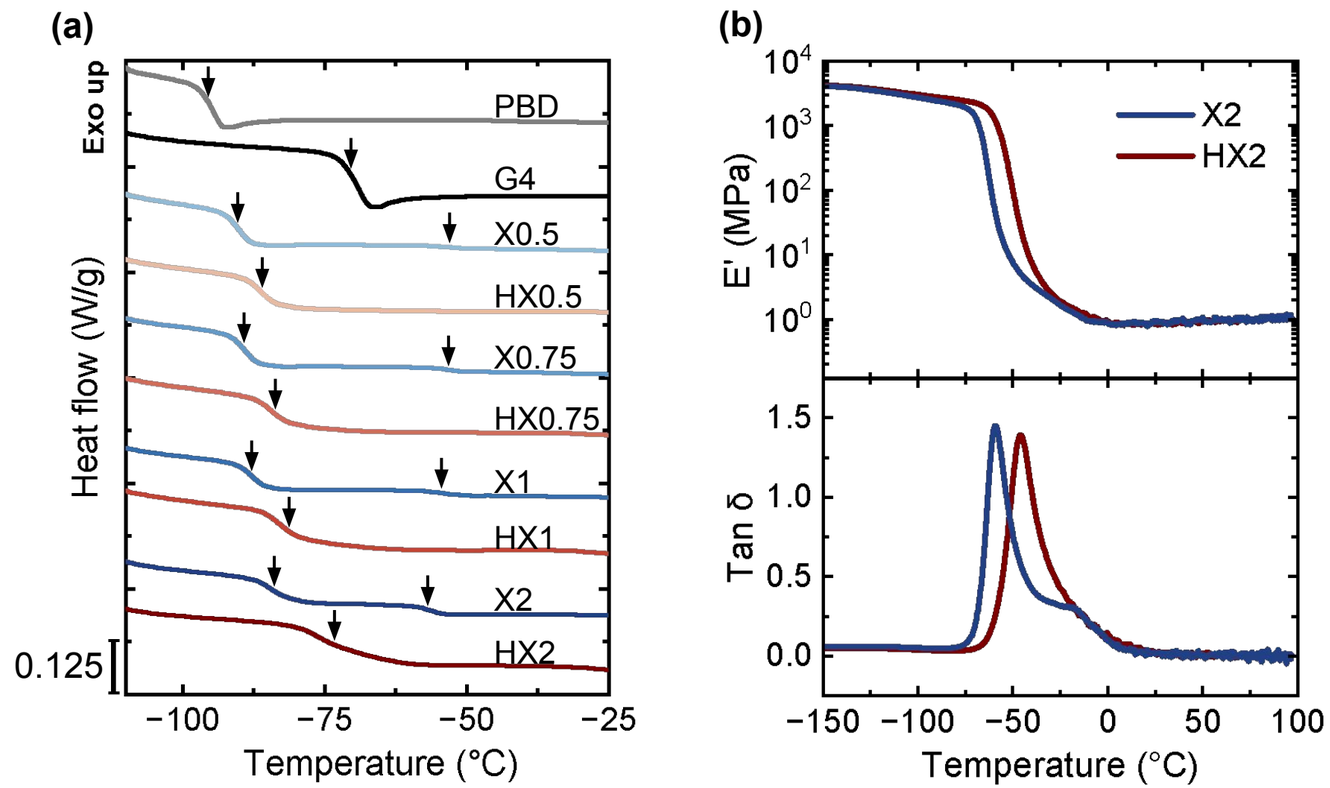


Figure 3: (a) DSC thermograms of neat PBD, neat G4, and photocured PBD-G4 samples with (X0.5-X2) and without (HX0.5-HX2) macroscopic phase separation at varying compositions; **(b)** DMA temperature sweeps of X2 and HX2: *E' vs. T* (top) and tan *δ vs. T* (bottom).

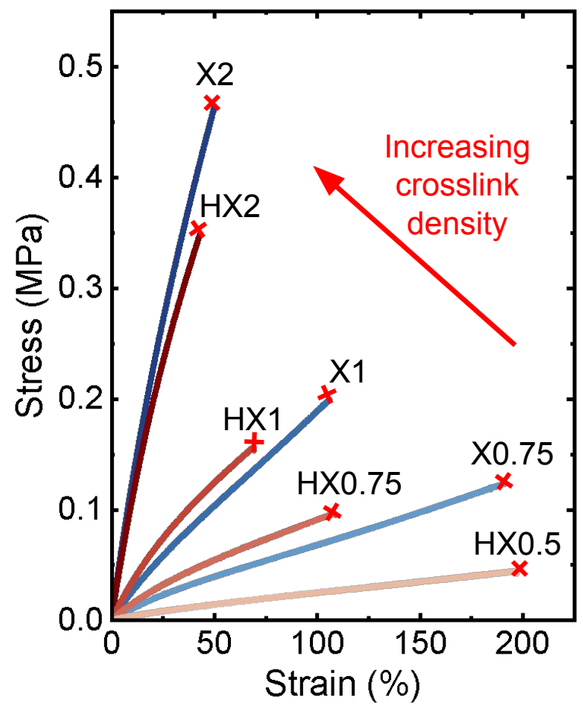


Figure 4: Room-temperature tensile test results for photocured PBD-G4 samples with (X0.75-X2) and without (HX0.5-HX2) macroscopic phase separation at varying compositions. Note that X0.5 film was sticky and not self-standing, thus not suitable for tensile testing.

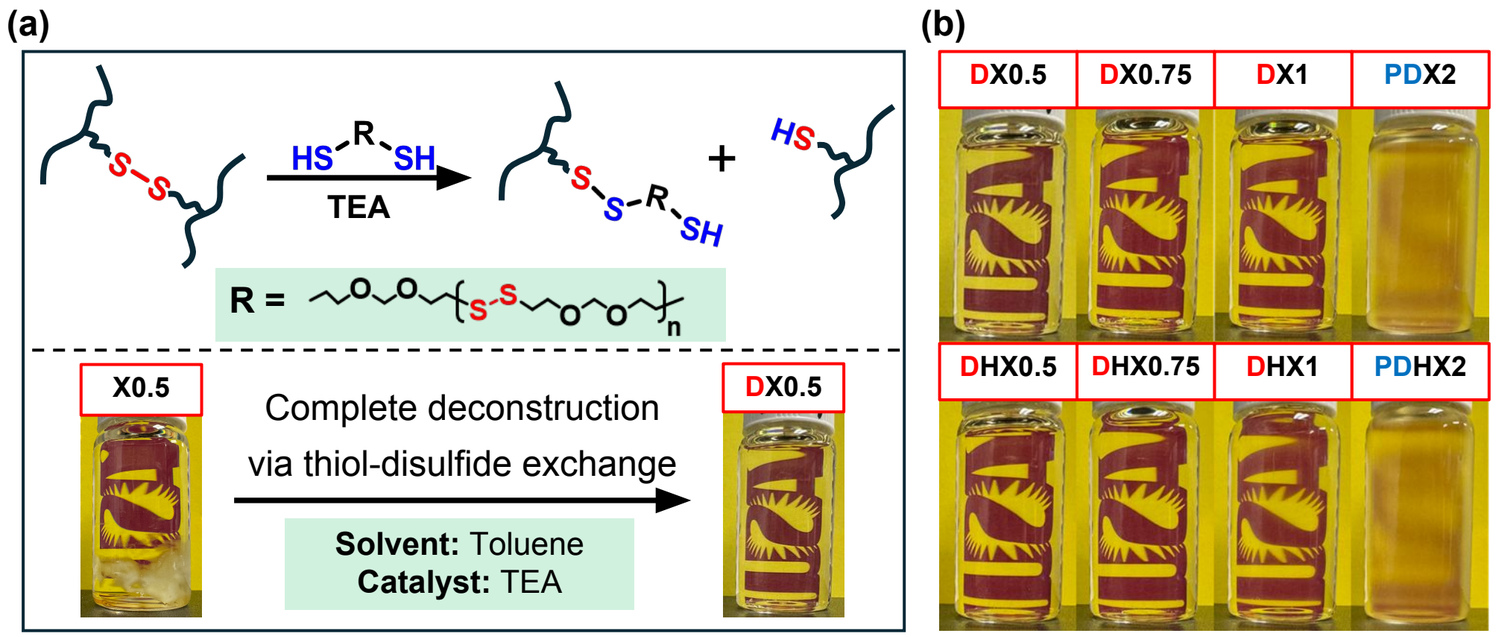


Figure 5: (a) Reaction scheme and decrosslinking experiment showing the deconstruction of thiol-ene PBD-G4 networks through dynamic thiol-disulfide exchange reactions; **(b)** Decrosslinking results for photocured PBD-G4 samples with (X0.5-X2) and without (HX0.5-HX2) macroscopic phase separation at varying compositions. **D** represents for complete network decrosslinking, while **PD** represents for partial network decrosslinking.

Table 1: Compositions of the phase-separated (X series) *vs.* phase-mixed (HX series) PBD-G4 photoresin formulations employed in this study, which contain varying PBD: G4 molar ratios.

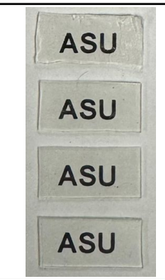
Sample	Molar Ration [PBD]: [G4]	Mass Ratio PBD: G4	Photo-initiator	THF	Visual appearance
X0.5	1.00: 0.50	5.00: 0.49	1 wt%	/	
X0.75	1.00: 0.75	5.00: 0.735			
X1	1.00: 1.00	5.00: 0.98			
X2	1.00: 2.00	5.00: 1.96			
HX0.5	1.00: 0.50	5.00: 0.49	1 wt%	250 wt%	
HX0.75	1.00: 0.75	5.00: 0.735			
HX1	1.00: 1.00	5.00: 0.98			
HX2	1.00: 2.00	5.00: 1.96			

Table 2: Gel fractions, thermal and mechanical properties of cured PBD-G4 samples with (X0.5-X2) and without (HX0.5-HX2) macroscopic phase separation at varying compositions.

Sample[a]	Gel fraction [%]	G′ [MPa] [b]	Young Modulus[c] [MPa]	Tensile Strength[c] [MPa]	Elongation at break[c] [%]
X0.5	-90, -53	0.005	/	/	/
X0.75	-89, -53	0.02			
X1	-88, -54	0.07			
X2	-84, -57	0.31			
HX0.5	-86	0.007			
HX0.75	-84	0.03			
HX1	-81	0.08			
HX2	-73	0.14			

[a]Reported errors are standard deviations from multiple measurements.
[b] G′ measured by rheometer at 0.1 rad s^{-1}
[c]Tensile properties measured at room temperature

Supporting Information

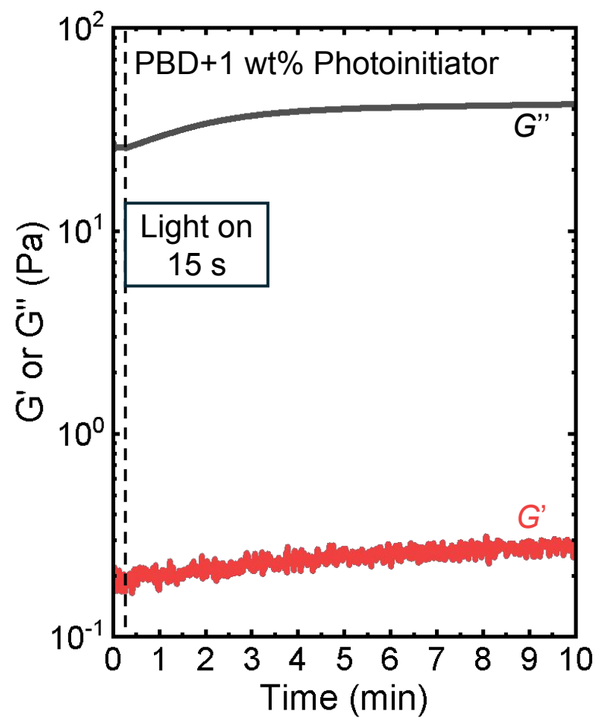


Figure S1: Photo-rheological analysis of the pure PBD (containing 1 wt% photoinitiator) under UV irradiation (i.e., 10 mW/cm^2), showing the evolution of G′ and G″ over time.

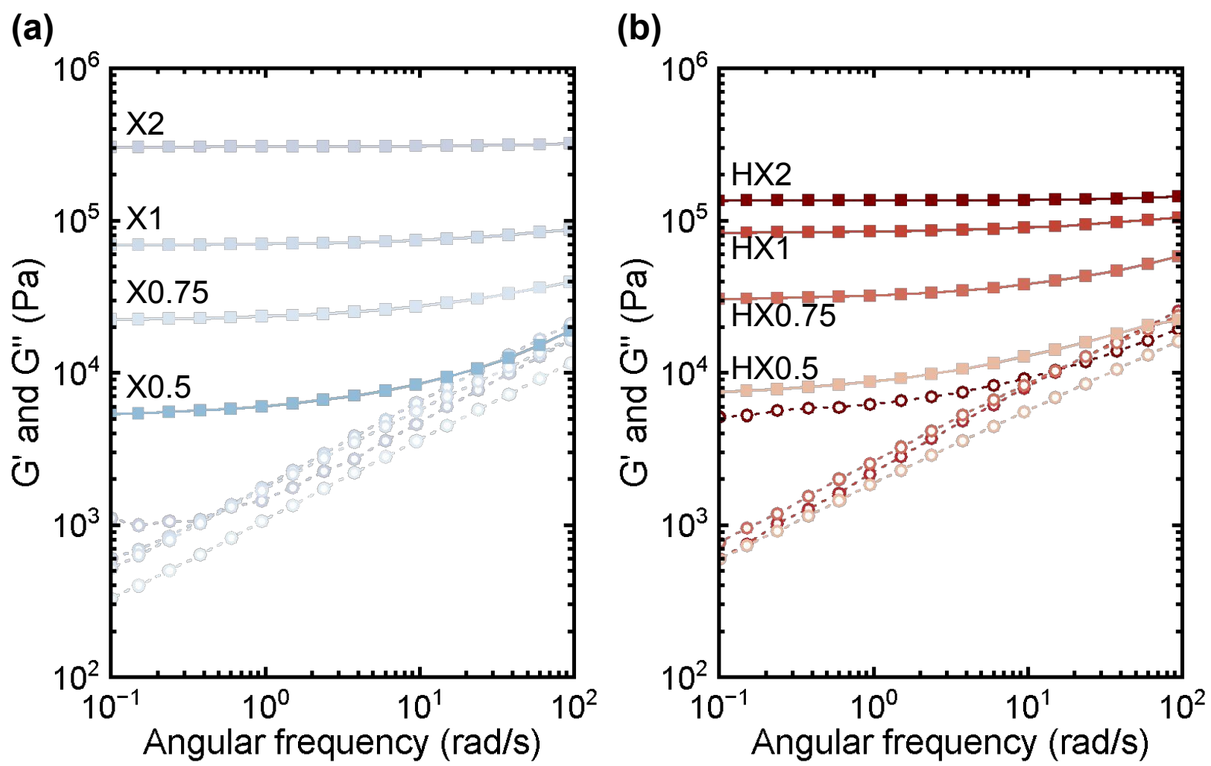
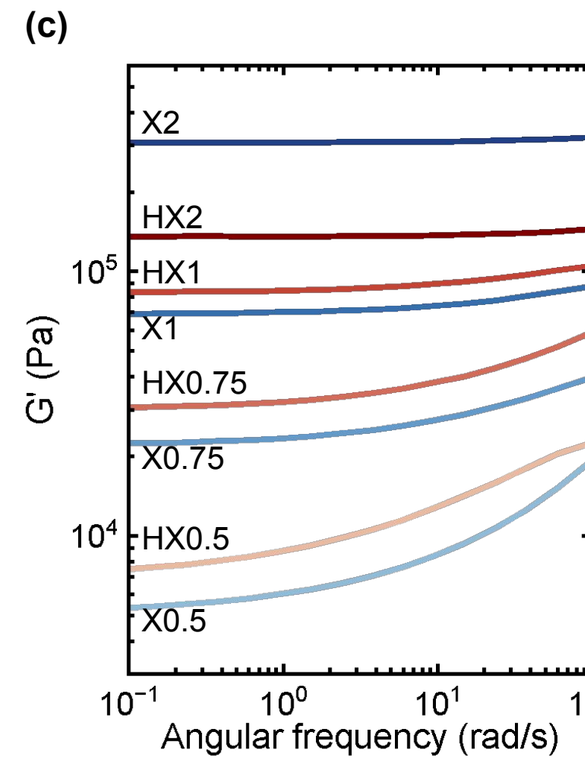
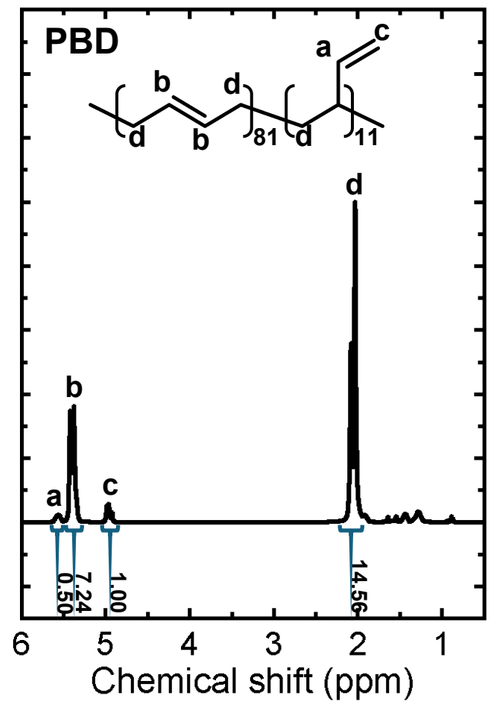


Figure S2: Room-temperature oscillatory rheological frequency sweep results for cured PBD-G4 networks.(a) Phase-separated X0.5-X2 and (b) single-phase HX0.5-HX2 samples. Solid symbols with solid lines denote the storage modulus (G′), and open symbols with dashed lines denote the loss modulus (G″).



(c) Direct comparison of the storage modulus (G′) between phase-separated (X) and single-phase (HX) networks at varying compositions.



Based on ^{1}H NMR,

b : c ≈ 7.24: 1.00

→ [1,4] : [1,2] ≈ 7.24 : 1.00

[1,2] ≈ 12.1%

[1,4] ≈ 87.9%

$n_{[1,4]}$ = (7.24/8.24)/(5000/54.09) ≈ 81

$n_{[1,2]}$ = (1.00/8.24)/(5000/54.09) ≈ 11

Figure S3: ^{1}H NMR spectrum of PBD in chloroform-d.

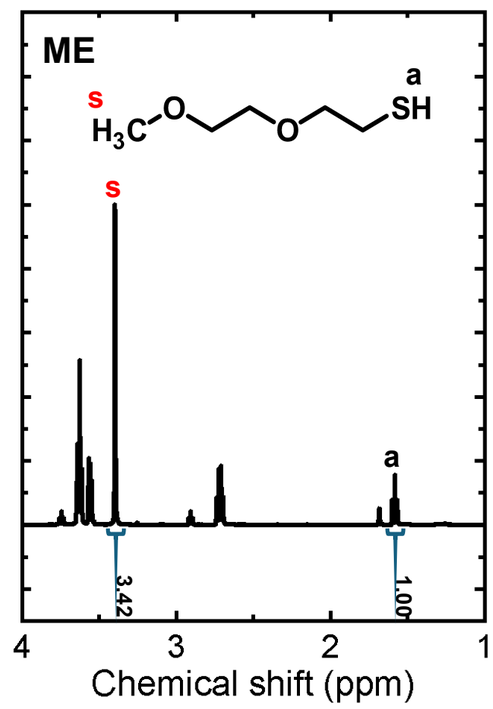


Figure S4: ^{1}H NMR spectrum of ME in chloroform-d.

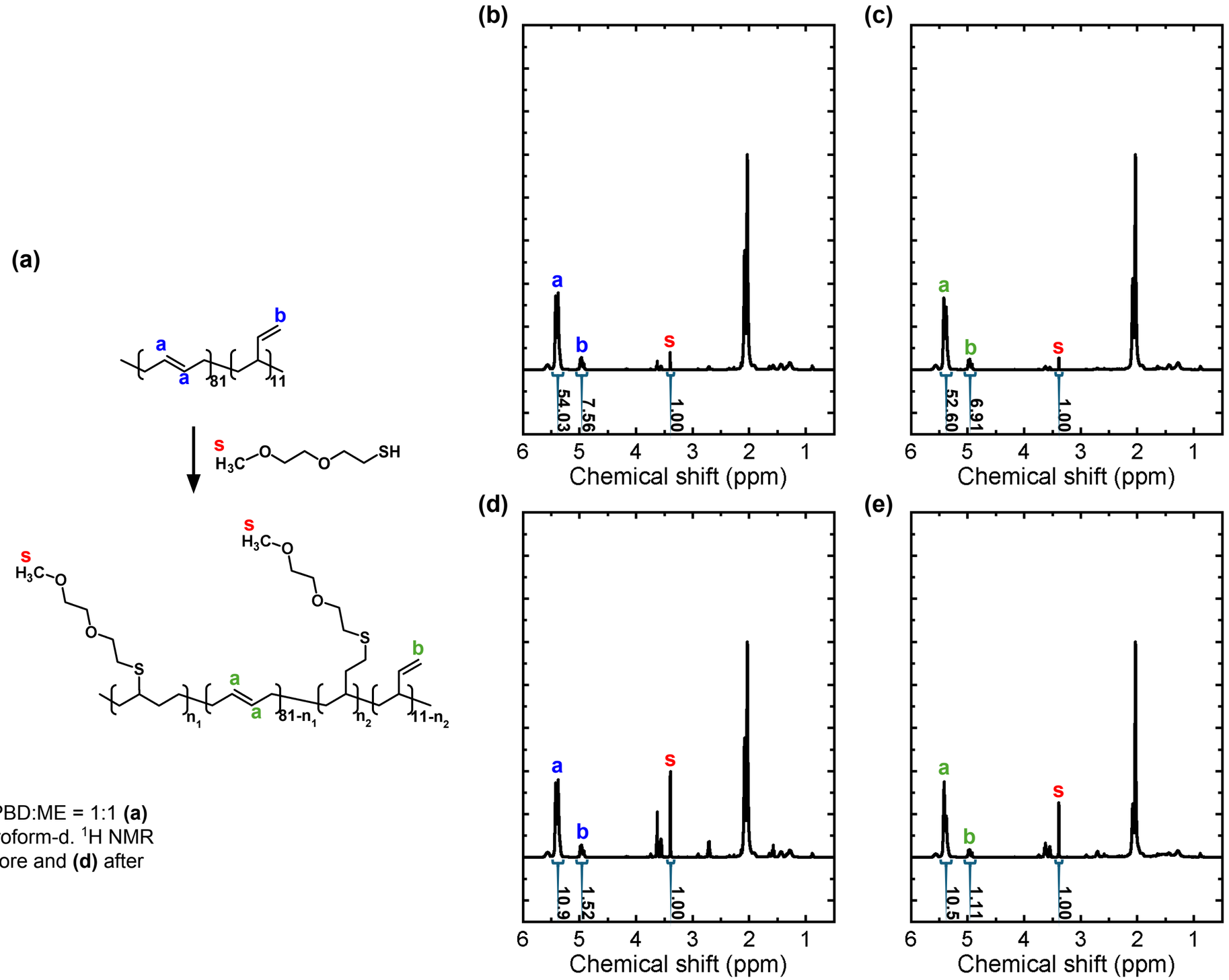


Figure S5: ^{1}H NMR spectrum of PBD:ME = 1:1 **(a)** before and **(b)** after curing in chloroform-d. ^{1}H NMR spectrum of PBD:ME = 1:5 **(c)** before and **(d)** after curing in chloroform-d.

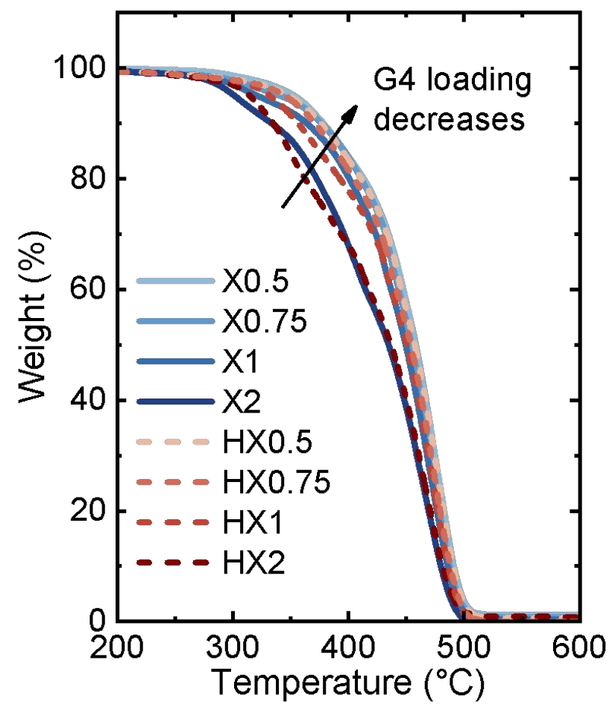


Figure S6: TGA spectrum of phase-separated and homogeneous samples.